CEE 298 Simulation Testing Project Scenario Document

1. Introduction

This document lists meaningful longtail scenarios that can serve as key use case scenarios for testing the performance of ADS agents, particularly with cooperative sensing capabilities.

2. Testing Scenarios

In the simulation environment, CARLA will be used to construct testing scenarios for those use cases with appropriate traffic conditions. Based on our previous project experience, the OpenCDA's simulation platform, which is built on CARLA, will be used to implement the planning module. The scenarios covering all the use cases listed in the proposal are discussed in this section.

2.1 Scenario A: Using CP to respond to potential red-light violators proactively

2.1.1. Motivation and Scenario Description

This scenario is used to evaluate the use of cooperative perception to proactively identify and communicate red-light violation risk when a connected automated vehicle (CAV) cannot directly perceive the traffic signal due to occlusion or limited visibility. The occlusion can come from other large vehicles (i.e., trucks or buses) blocking the viewing angle of the ego CAV. The limited visibility can come from road geometry characteristics, such as the location of parking or large vegetation blocking the traffic light leading to line of sight of issue. In this situation, other CAVs with clear visibility of the traffic light perceive the signal state and the motion of nearby vehicles, predict the future trajectory of nearby vehicles, and assess whether the CAV with limited visibility is approaching the intersection at a speed that creates a high risk of running a red light, as shown in the Figure 1.

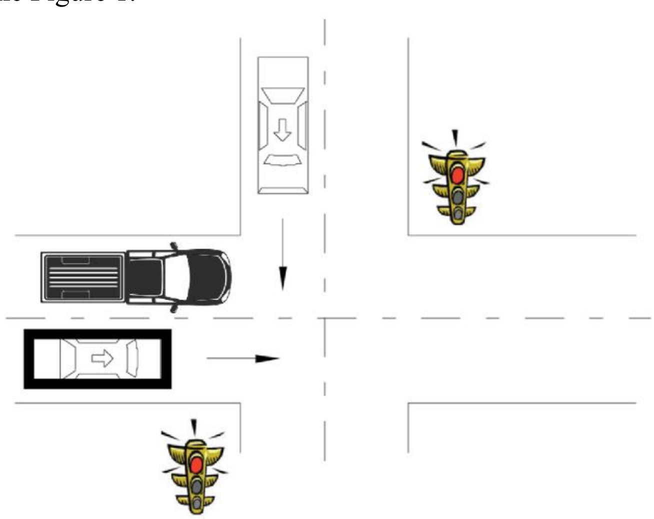

Figure 1. Red light violation scenario. Vehicles with limited signal visibility receive shared signal state, surrounding traffic detections, and motion predictions from other connected vehicles to assess red-light violation risk at the intersection.

When a high risk is identified, the perceiving CAVs transmit cooperative perception messages to the at-risk vehicle. These messages convey the perceived traffic signal state, the estimated motion state of the at-risk vehicle, and an assessment of red-light violation risk. The receiving CAV incorporates this information into its planning process to adjust its approach behavior and reduce the likelihood of an intersection conflict.

In simulation, this scenario is implemented using OpenCDA with CARLA and SUMO. One CAV is configured with limited or occluded traffic-light perception, while another CAV maintains clear visibility of the signal. Cooperative perception messaging is enabled between vehicles. The ego vehicle is evaluated under both local-only perception and CP-enabled conditions to assess the impact of cooperative perception on planning behavior.

In field testing, similar to the simulation test, one vehicle experiences traffic-light occlusion caused by geometry or surrounding traffic, while another vehicle maintains direct visibility of the signal. Vehicles exchange cooperative perception messages in real time. The ego vehicle's planning response to red-light violation risk is observed under CP-enabled operation.

2.1.2. Scenario Variants & Parameterization

- **Traffic signal visibility conditions.** They are varied by introducing different sources of occlusion, including partial or full blockage by large vehicles (e.g., trucks or buses) and reduced visibility caused by road geometry, such as curved approaches or offset signal placement. The duration and severity of signal occlusion are adjusted to represent both transient and sustained visibility loss.
- **Vehicle approach dynamics.** They are parameterized by varying the ego vehicle's approach speed, distance to the intersection at the onset of occlusion, and deceleration behavior, which influence the assessed risk of red-light violation and allow evaluation across low-, medium-, and high-risk approach conditions.

2.2 Scenario B: Using CP to Navigate Difficult Geometric Conditions Affecting Line-of-Sight

2.2.1. Motivation and Scenario Description

This scenario is used to evaluate the use of cooperative perception to support vehicle planning in roadway environments where difficult geometric conditions limit line-of-sight and result in incomplete or delayed perception of surrounding road users. These conditions can arise from both sharp horizontal and sharp vertical curves, commonly found on rural two-lane roads or highways, where tight bends or crest curves limit sight distance and prevent early detection of opposing vehicles, crossing traffic, cyclists, or pedestrians until the vehicle reaches the curve apex.

Difficult visibility conditions also occur in dense urban environments, locations with poor parking layouts, and T-intersections, where parked vehicles, buildings, roadside structures, or constrained intersection geometry block direct line-of-sight. In urban curbside parking scenarios, large parked vehicles such as vans, trucks, or SUVs may occlude cross traffic, pedestrians, or cyclists. At T-intersections, limited approach angles and building setbacks can restrict visibility of through traffic. Additional visibility constraints may arise at skewed or offset intersection approaches and locations with fixed roadside structures such as retaining walls, bridge abutments, barriers, or dense vegetation.

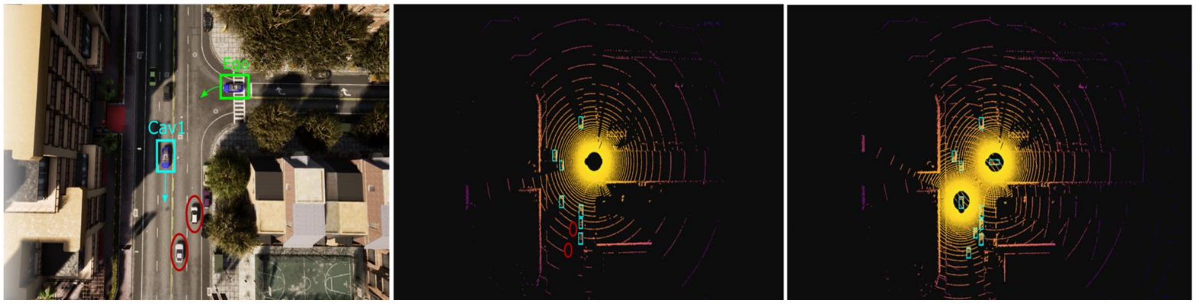

Figure 2. Occlusion Scenario. The ego vehicle is at the T-intersection suffering occlusion issue due to the vegetation. With cooperative perception, the CAV1 can share road vehicles (two cars in the red circle) information with the ego vehicle.

In this scenario, a CAV approaching the roadway segment or intersection relies on onboard sensors that provide only partial situational awareness due to geometric and environmental constraints. As a result, the ego vehicle may detect other vehicles or vulnerable road users late, leaving limited time for safe planning responses. Other CAVs positioned upstream, downstream, or on intersecting approaches maintain clearer views of the roadway and are able to observe road users that are partially or fully occluded from the ego vehicle's perspective. These perceiving CAVs track the positions and motions of nearby road users, predict their future trajectories, and transmit cooperative perception messages to the ego vehicle. The shared information extends the ego vehicle's situational awareness beyond its local sensing limits and enables earlier and more informed planning under constrained visibility conditions. As shown in Fig. 2, cooperative perception aggregates complementary sensor views from multiple CAVs, improving detection coverage of occluded road users compared to onboard sensing alone.

In simulation, this scenario is implemented using OpenCDA. Road networks are configured with challenging geometric features that restrict line-of-sight, including curved road segments, dense urban blocks with on-street parking, and T-intersections. The ego vehicle is placed in a location with limited visibility, while one or more neighboring CAVs maintain clearer perception of the roadway. Cooperative perception messaging is enabled between vehicles, and the ego vehicle is evaluated under both local-only perception and CP-enabled conditions to assess the impact of cooperative perception on planning behavior.

In field testing, similar to the simulation test, the scenario is deployed on road segments or intersections with known geometric visibility limitations. One vehicle experiences restricted line-of-sight due to road geometry or surrounding structures and other vehicles maintain better views of the roadway. Vehicles exchange cooperative perception messages in real time. The ego vehicle's planning response under CP-enabled operation will be evaluated.

2.2.2. Scenario Variants & Parameterization

- **Road geometry configurations**. They are varied by introducing curved road segments (horizontal and vertical curves), skewed or offset intersections, T-intersections, dense urban street layouts, and locations with roadside structures that restrict line-of-sight.
- **Parking-related occlusion conditions**. They are parameterized by varying the presence, size, and placement of parked vehicles along curbs, including scenarios with poor parking layouts that block visibility of cross traffic or pedestrians.
- **Visibility constraints**. Except for road geometry, they are parameterized by adjusting the severity and duration of line-of-sight limitations caused by relative vehicle positioning.
- **Vehicle motion states**. They are parameterized by varying ego and surrounding vehicle speeds and relative spacing to represent different interaction dynamics under constrained visibility.

2.3 Scenario C: Using CP to inform vehicles about VRUs at mid-block crossings or intersections, VRUs jaywalking, VRUs occluded by large vehicles, and VRUs sharing the right-of-way with cars

2.3.1. Motivation and Scenario Description

This scenario is used to evaluate the use of CP to improve vehicle awareness of vulnerable road users (VRUs) in situations where onboard sensing alone may be insufficient due to occlusion, limited visibility, or complex interaction dynamics. The scenario focuses on VRUs at mid-block crossings or intersections, VRUs jaywalking, VRUs occluded by large vehicles, and VRUs sharing the right-of-way with vehicles. In these situations, a CAV may have difficulty detecting pedestrians or cyclists in a timely manner because VRUs are partially or fully blocked by parked vehicles, buses, trucks, roadside structures, or other traffic participants. VRUs may also enter the roadway unexpectedly, such as during jaywalking or informal mid-block crossings, creating sudden risks of conflict.

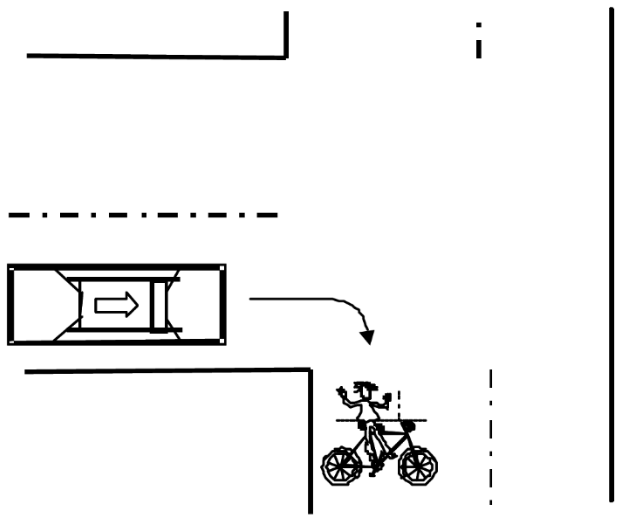

Figure 3. VRU Scenario. A cyclist enters the vehicle path from an occluded region near a mid-block crossing or intersection. Cooperative perception enables earlier detection and motion awareness of VRUs by sharing observations from vehicles with clearer viewpoints, supporting safer planning and yielding behavior.

At intersections or shared spaces, VRUs and vehicles may have overlapping right-of-way, requiring accurate perception and anticipation of VRU motion. As illustrated in Figure 3, VRUs may enter the vehicle path from occluded regions, making timely detection challenging with onboard sensors alone. Other CAVs with clearer viewpoints are able to detect VRUs earlier and track their motion. These perceiving CAVs estimate the positions and movement trajectories of VRUs and share cooperative perception messages with nearby vehicles. The shared information enables the ego vehicle to become aware of VRUs that are occluded or outside its immediate sensing range. The receiving CAV incorporates cooperative perception information into its planning process to adjust speed, trajectory, or yielding behavior, with the goal of reducing collision risk and improving safety in VRU-heavy environments.

In simulation, we will still use OpenCDA to implement the scenario. VRUs are introduced at mid-block crossings and intersections, including cases where VRUs are occluded by large vehicles or enter the roadway unexpectedly. One or more CAVs maintain clear visibility of the VRUs, while the ego vehicle experiences partial or full occlusion. Cooperative perception messaging is enabled between vehicles.

In field testing, similar to the simulation setup, the scenario is deployed in urban or suburban environments with frequent pedestrian or cyclist activity. One vehicle experiences limited visibility of VRUs due to surrounding traffic or roadside conditions and other vehicles maintain clearer views. Cooperative perception messages are exchanged in real time, and the ego vehicle's planning response to VRU-related risks is observed under CP-enabled operation.

2.3.2. Scenario Variants & Parameterization

- **VRU types and behaviors.** They are varied by introducing pedestrians and cyclists with different movement patterns, including crossing at designated mid-block crossings or intersections, jaywalking across traffic lanes, and moving parallel to traffic while sharing the right-of-way with vehicles.
- **VRU visibility conditions.** They are varied by introducing partial or full occlusion of VRUs caused by parked vehicles, buses, trucks, roadside structures, or other traffic participants, as well as by adjusting the duration and timing of occlusion.
- **VRU entry timing and location.** They are parameterized by varying the time and position at which VRUs enter the roadway, including sudden appearance from behind large vehicles or roadside obstacles.

2.4 Scenario D: Using CP to inform vehicles about a foreign object in the roadway

This scenario is used to evaluate the use of cooperative perception to inform vehicles about the presence of foreign objects in the roadway that may not be detected in a timely manner using onboard sensing alone. Foreign objects may include debris, fallen cargo, disabled vehicles, roadway hazards, or other unexpected obstacles that pose safety risks to approaching traffic. The foreign objects will be included in the Scenarios E and F within the work zone situations. Figure 4 illustrates an example of a foreign object located along the vehicle's intended path that is partially or fully occluded from direct view, where upstream vehicles or roadside infrastructure detect the hazard and share its location through cooperative perception to support earlier awareness and safer maneuver planning.

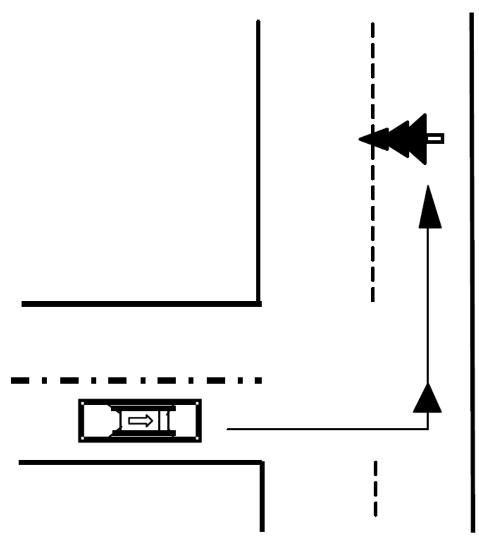

Figure 4. Foreign object scenario. Foreign objects located along the intended travel path are detected by upstream vehicles or infrastructure and shared to support early hazard awareness and safe maneuver planning.

2.5 Scenario E: Using CP to Inform Vehicles About Roadway Hazards and Support Passing and Merging Maneuvers

2.5.1. Motivation and Scenario Description

This scenario is used to evaluate the use of cooperative perception to support vehicle planning in work zone environments, where temporary roadway configurations and the presence of foreign objects related to construction activities create complex and potentially hazardous conditions. Foreign objects may include fallen construction materials, displaced cones or barrels, equipment components, or temporarily stopped construction vehicles within or near active travel lanes.

In this scenario, a CAV approaching a work zone may have limited ability to perceive the full work zone layout and the presence of foreign objects using onboard sensors alone. Occlusion by construction equipment, barriers, large vehicles, or roadway curvature may delay detection of lane closures, lane shifts, or foreign objects within the travel path. As a result, the ego vehicle may encounter sudden merging, passing, or obstacle-avoidance requirements with limited time for safe planning. Other CAVs traveling upstream of the work zone or already within the work zone may maintain clearer views of temporary traffic control elements and foreign objects on the roadway. These perceiving CAVs detect work zone features, identify foreign objects within or adjacent to travel lanes, track surrounding vehicle motion, and share cooperative perception messages with nearby vehicles. The shared information conveys the location of lane restrictions and the presence of foreign objects. The receiving CAV incorporates cooperative perception information into its planning process to adjust speed, lane position, and trajectory in advance of the work zone. Cooperative perception enables earlier preparation for merging, passing, or obstacle avoidance, reducing the likelihood of abrupt maneuvers and improving safety in work zone environments. As shown in Figure 5, vehicles receive early information about lane shifts and work zone constraints, enabling timely lane changes and speed adjustments before reaching the coned-off area.

Figure 5. Passing and merging scenario.

In simulation, this scenario is implemented using OpenCDA. Work zone configurations are introduced on roadway segments, including lane closures, lane shifts, and narrowed lanes, along with foreign objects placed within or near active lanes. One or more CAVs operate within or upstream of the work zone and share cooperative perception messages, while the ego vehicle evaluates planning behavior under both local-only perception and CP-enabled conditions.

In field testing, similar to the simulation setup, the scenario is deployed in controlled or active work zone environments. Foreign objects related to construction activities are safely represented using controlled test artifacts. One vehicle detects work zone features and foreign objects ahead of the ego vehicle and transmits cooperative perception messages in real time. The ego vehicle's planning response to work zone constraints and foreign objects is evaluated under CP-enabled operation.

2.5.2. Scenario Variants & Parameterization

- **Work zone configurations**. They are varied by introducing single-lane or multi-lane closures, lane shifts, and narrowed lanes.
- **Foreign object characteristics**. They are parameterized by varying the type, size, and placement of construction-related foreign objects within or adjacent to travel lanes.

2.6 Scenario F: Awareness for high-level path/route planning needs

2.6.1. Motivation and Scenario Description

This scenario is used to evaluate the use of cooperative perception to support vehicle planning in work zones and emergency events, where roadway conditions are disrupted and onboard sensing alone may not provide sufficient situational awareness. These environments often involve temporary lane closures, lane shifts, foreign objects, stopped vehicles, emergency responders, and dynamic traffic control, creating elevated safety risks for both road users and first responders.

In work zone and emergency response environments, a CAV approaching the affected area may experience severely limited visibility of downstream conditions due to occlusions caused by large vehicles, construction equipment, emergency vehicles, temporary barriers, or altered roadway geometry. These occlusions can prevent the ego vehicle from reliably detecting lane closures, lane shifts, foreign objects, stopped or slow moving vehicles, or the presence and activity of emergency responders using onboard perception alone. Under such conditions, other CAVs operating upstream, within, or downstream of the work zone or emergency scene may maintain clearer views of the affected area. These perceiving CAVs are able to identify work zone layouts, cone defined lane restrictions, roadway hazards, emergency vehicles, and responder motion, and generate cooperative perception messages that communicate the location, extent, and dynamics of these critical elements.

Once cooperative perception messages are received, the ego CAV incorporates the shared information into its planning and decision-making processes to adjust its behavior in advance. The cooperative perception information can be used to **update short term planning decisions**, such as speed control, lane selection, and local trajectory generation, as well as **longer term planning decisions** at the global level. When the work zone or emergency event affects roadway availability or route feasibility, the ego CAV may revise its global route to avoid constrained or unsafe segments while simultaneously adapting its local trajectory. By supporting coordinated updates to both short term vehicle motion and long term route planning, cooperative perception enables safer and more efficient navigation through work zone and emergency environments while enhancing the safety of first responders and road users.

As shown in Figure 6a, cooperative perception is enabled by upstream CAVs that maintain clearer visibility of the work zone or emergency scene despite occlusions affecting the ego vehicle. These perceiving vehicles detect critical elements such as the cone-defined lane closure, blocked roadway segments, and the presence of a stopped police vehicle, and share this information through cooperative perception messages. By receiving these messages, the ego CAV gains advance awareness of downstream lane restrictions and hazards that are not yet observable through local sensing, allowing it to proactively adapt its planning and control before entering the affected area.

Meanwhile, as shown in Figure 6b, cooperative perception enables global route adaptation in response to emergency vehicle activity that is not observable to the ego CAV due to distance and intervening roadway geometry. In this scenario, a single upstream CAV has direct visibility of an emergency response scene involving an ambulance and cone-defined road closures that restrict access through the intersection. Because this disruption is out of sight and difficult to detect for the ego vehicle, the upstream CAV’s closer vantage allows it to recognize the emergency operation and resulting roadway blockage and share this information through a cooperative perception message. Upon receiving the message, the ego CAV incorporates the shared information into its global planning process and proactively updates its route to avoid the affected roadway segment. This capability supports safer navigation, reduces unnecessary delays, and minimizes interference with emergency response activities.

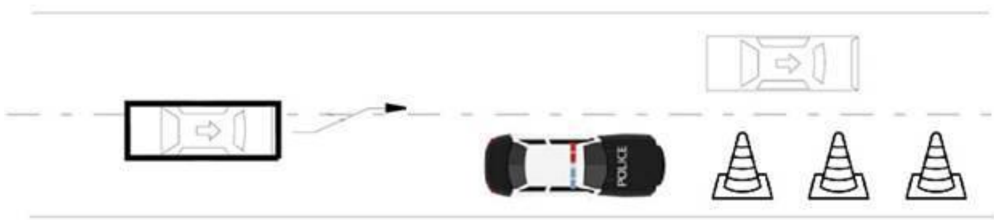

Figure 6a. High-level route planning scenario. The upstream CAVs detect lane closures, roadway obstructions, and emergency vehicles and share this information with an occluded downstream CAV to support proactive planning and safer navigation.

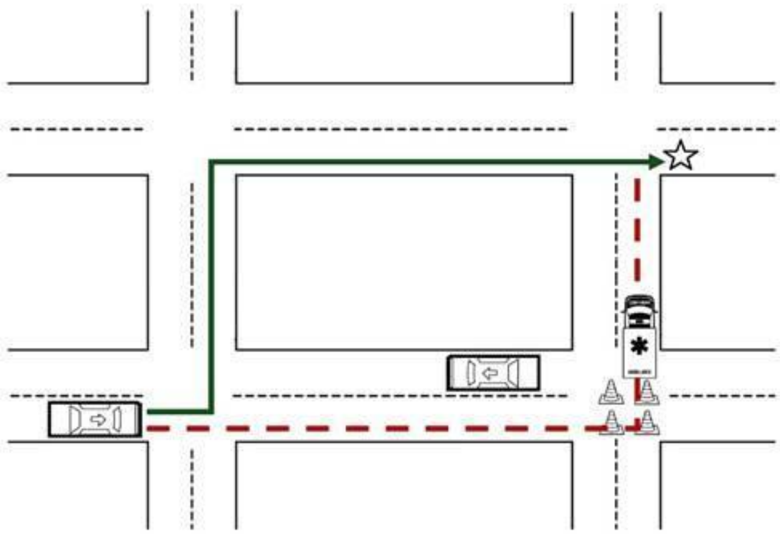

Figure 6b. Cooperative perception–enabled global route update, where the ego CAV reroutes based on shared information about an out-of-sight emergency scene, with the red dotted line representing the original route and the green solid line representing the updated route.

In simulation, this scenario is implemented using OpenCDA. Work zone layouts and emergency event configurations are introduced on roadway segments, including lane closures, lane shifts, foreign objects, and stopped emergency vehicles. One or more CAVs detect these conditions and share cooperative perception messages, while the ego vehicle evaluates planning behavior under both local-only perception and CP-enabled conditions.

In field testing, similar to the simulation setup, the scenario is deployed in controlled work zone or emergency response environments. Foreign objects and responder presence are represented using controlled test artifacts and vehicles. CAVs exchange cooperative perception messages in real time, and the ego vehicle's planning response is evaluated under CP-enabled operation.

2.6.2. Scenario Variants & Parameterization

- **Operational context**. They are varied by deploying active work zones, emergency response scenes, or combined work zone–emergency situations.
- **Lane restriction configurations**. They are parameterized by varying lane closures, lane shifts, and reduced lane widths.
- **Foreign objects and roadway hazards**. They are varied by introducing construction-related debris, equipment, or disabled vehicles within or near active lanes.
- **First responder presence**. They are parameterized by varying the number, location, and movement patterns of emergency responders and emergency vehicles.

Initial Insights from Steering Committee Interviews

1. High-Priority Safety Scenarios

These represent the most consistent areas of consensus across OEMs, ADS developers, and DOTs:

- **Vulnerable Road Users (VRUs):** Detecting pedestrians, e-bikes, and scooters, especially those occluded by parked cars or moving between vehicles. High-speed VRUs (e-scooters) are a major concern due to shortened decision-making windows.
- **Occlusions & Blind Spots:** "Seeing" around corners at difficult road geometries, unprotected left turns, and entering major roads from minor streets where the view is blocked by parked cars or infrastructure.
- **Red-Light Violations:** Using infrastructure to detect and warn of vehicles likely to run a red light (Intersection Movement Assist).
- **Emergency Vehicles:** Providing early detection and directionality of emergency vehicles (sirens/lights) to assist in route planning and clearing paths.

2. Strategic & Operational Applications

Beyond immediate collision avoidance, stakeholders identified long-term operational uses:

- **ODD Validation & Expansion:** Using infrastructure data as a "ground truth" to compare against vehicle perception, helping fleets decide if a new area (Operational Design Domain) is safe to enter.
- **Work Zone Safety:** Real-time communication of vehicle intrusions into work zones and the status of work-zone equipment, moving beyond static lane-closure plans.
- **Strategic Map Updates:** Identifying "world map" anomalies (e.g., unexpected debris, new road markings, or stalled vehicles) rather than just maneuver-level objects.
- **Queue & Speed Management:** Monitoring queue buildup and providing curve speed warnings, particularly for heavy trucking with long (11–12s) headway requirements.